MARK A. MATHESON

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SUMMARY

Automation and controls specialist with 15+ years commissioning, programming, and troubleshooting industrial robotic systems in high-uptime production. Has brought new robot cells online from install through production handoff for ABB and BMW, and owns fault recovery across a mixed ABB, Dürr, and Fanuc fleet. Works hands-on across controllers, PROFINET communication, electrical, pneumatic and fluid systems, sensors, and vision, and leads issues to closure with technicians and engineering. Comfortable with field travel, shift and weekend work, and fast build-test-fix cycles.

CORE SKILLS

Commissioning & Integration: New robot cells (sealer, paint, pick & place), IRC5/IRC5P controllers, IPS paint systems, vision guidance, BMW IPP and Wax line commissioning team

Controls & Programming: Robot-to-PLC interface troubleshooting (PROFINET, I/O handshakes), ABB RAPID / RobotStudio, Dürr 3D Onsite, Fanuc dispense, Siemens Process Simulate, ABB RobView operator screens and dashboards, offline programming and hot edits; learning Siemens TIA Portal

Electrical & Hardware: IRC5P electrical (ABB certified), electromechanical and servo/brake repair, sensors and vision systems, pneumatic and fluid delivery systems

Safety Systems: ABB SafeMove Pro (certified) and SafeMove 2 (trained), lockout/tagout (BMW and ABB trained, daily practice), e-stops and safety circuits, work in zero-failure safety-critical environments

Industrial Networking & Linux: Network configuration (VLANs, switching), Linux, Proxmox virtualization; CCNA in progress

Troubleshooting & RCA: Fault diagnosis under production pressure, downtime recovery, issue tracking to closure

Vision Systems: ABB Flex Vision, ISRA Vision, VMT robot guidance setup and troubleshooting

Tools & Scripting: WinSCP/FileZilla scripted robot backups, Autodesk Fusion and Bambu Studio for 3D-printed tooling, AI-assisted scripting (Claude Code)

Fabrication & Documentation: CAD and 3D-printed custom tooling, technical procedures, engineering field changes, technician training

PROFESSIONAL EXPERIENCE

Robot Specialist | BMW Manufacturing, Greer, SC *Feb 2023 – Present*

- Own technical development of paint shop automation: robot programming for new models, design changes, and continuous improvement across ABB, Dürr, and Fanuc systems.
- First responder for planned and unplanned downtime: diagnose controller, electrical, pneumatic, and process faults and restore production, sustaining 95% equipment availability across the paint robot fleet.
- Troubleshoot robot-to-PLC communication with controls engineers: monitor I/O and PROFINET traffic, chase device drops, and isolate faults to the failing component. Isolated an intermittent fault that ran hours to days between drops: used RobotStudio monitoring, NIC link status, and the boot console to clear the IPS Process Interface Board, then traced it to a failing LAN-side switch inside the IRC5 that dropped the board while the HMI network stayed up.
- Hold 60-second station cycle times through path optimization and parallel sequencing, such as firing vision measurement and computing corrections during a tool change.
- Validate new materials and process changes in the robot lab before release to production.
- Track and drive issues to closure with engineering, maintenance, and production teams.
- Automated weekly backups for ~40 ABB robots: controller tasks plus a WinSCP script that pulls each backup to network storage, verifies it, and keeps the three newest on the robot, saving 100+ labor hours a year.
- Design and 3D print custom robot tooling in CAD for fast, low-cost fixes.

Field Service Engineer | Medtronic *Jan 2021 – Feb 2023*

- Installed, maintained, and upgraded life-critical ventilation systems (Puritan Bennett 980/840) across a 32-account, three-state territory, working independently.
- Diagnosed faults across integrated hardware, pneumatics, sensors, and software in a zero-failure patient-safety environment.
- Executed engineering field changes and upgrades under controlled procedures with full documentation; trained customer technical staff onsite.

Field Service Engineer | ABB *Nov 2017 – Dec 2020*

- Commissioned new robot cells for sealer and paint applications at customer plants, from install through production handoff, including BMW's IPP4 paint line and its feather duster cell, and a Honda bumper paint line.
- Programmed and commissioned a 16-robot masking plug cell for BMW with vision guidance: launched on schedule at a 60-second cycle and 95% plug success; ABB Field Service Employee of the Month (June 2018).
- Integrated IRC5/IRC5P controllers, IPS paint systems, and Flex Vision guidance; programmed pick-and-place applications.
- Delivered cycle time improvements, robot brake repairs, and onsite troubleshooting using RobotStudio, RobView, and Shop Floor Editor.

Robot Programmer | BMW Manufacturing, Greer, SC *Aug 2011 – Nov 2017*

- Created paint programs for new models, ran trial bodies, and fine-tuned robot paths.
- Set up color tables and brush parameters for Bell RB1000 applicators; corrected paint defects at the source.
- Used simulation software for hot edits to minimize production impact.

Diesel Technician | Morgan Corporation *Sep 2010 – Jul 2011*

- Overhauled diesel equipment and completed field repairs to minimize equipment downtime.

EDUCATION & CERTIFICATIONS

- Associate Degree, Management & Diesel Technology, WyoTech, Blairsville, PA (2010)
- ABB: IRC5 Programming II (2020), SafeMove Pro (2019), Flex Vision (2019), IRC5P Electrical Service (2018), RobotStudio I (2018), IRC5P Integrated Process Solution (2018)
- Fanuc: DispenseTool Operation and Programming (2017)
- Medtronic: Puritan Bennett 980 and 840 Ventilator Service (2021)
- Cisco CCNA (in progress)